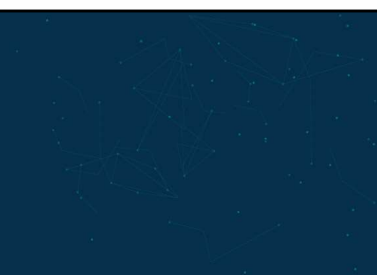


Attacking the Cloud

CHAPTER 6



6.01 - Cloud Attacks, Part 1

Episode **Cloud Attacks, Part 1**
title:

Objective: **3.4 Given a scenario, research attack vectors and perform attacks on cloud technologies**

CLOUD ATTACKS

- **Credential harvesting**
 - Using different techniques to compromise credentials used to access cloud resources
- **Privilege escalation**
 - Leveraging application or service vulnerabilities to carry out unauthorized actions

CLOUD ATTACKS

- **Account takeover**

- After compromising credentials, attacker changes access configuration to deny the rightful owner access

- **Metadata service attack**

- Service that provides information about how to access hosted services
- Compromised metadata service could report malicious service information

CLOUD ATTACKS

- **Misconfigured cloud assets**
 - Identity and access management (IAM)
 - Incorrect/malicious authentication/authorization
 - Federation misconfigurations
 - Deny access to authorized subjects or allow access to unauthorized subjects
 - Potential to leak credentials

CLOUD ATTACKS

- **Misconfigured cloud assets**
 - Object storage
 - Improper settings could allow unauthorized access
 - Containerization technologies
 - Improper container configurations can allow unauthorized connections
 - Could lead to further compromise

6.02 - Cloud Attacks, Part 2

Episode title: **Cloud Attacks, Part 2**

Objective: **3.4 Given a scenario, research attack vectors and perform attacks on cloud technologies**

CLOUD ATTACKS

- **Resource exhaustion**
 - Similar to classic
 - Denial-Of-Service (DoS) attacks
 - Distributed Denial of Service attack (DDoS) attacks
 - Threat to service and server availability

CLOUD ATTACKS

- **Cloud malware injection attacks**
 - Delivering malware to cloud-based VMs and containers
 - Similar to malware delivery for physical servers
- **Denial-Of-Service (DOS) attacks**
 - Saturate network or CPU with traffic/work to leave no capability for authorized processes

CLOUD ATTACKS

- **Side-channel attacks**

- Generally carried out by a malicious VM that sample system performance during cryptographic operations to disclose key information

- **Direct-to-origin attacks**

- Attacks that penetrate layers of devices intended to protect the true network's addresses
- Two-step attack – first find the real network, then attack it directly